MD TANVIR ALAM

Senior Software Engineer | Senior Backend Java | Agentic AI Platform

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GitHub: github.com/tanvir-ux | Portfolio: tanvir-ux.github.io | LinkedIn: linkedin.com/in/mdtanviralam
Open to: Remote from Bangladesh, or APAC (Singapore EP, Jakarta, India)

SUMMARY

Lead Software Engineer with 5+ years designing distributed systems, high-concurrency banking backends, and production Agentic AI platforms. Expert in Java, Spring Boot, Microservices, REST, Kafka, Kubernetes, Docker, AWS, PostgreSQL, Oracle, CI/CD, System Design, and Distributed Systems. Builds Agentic AI services with LangChain, RAG, and MCP (Model Context Protocol). Delivers core banking work on Oracle FLEXCUBE, Oracle Banking Payments (OBPM), and SWIFT for clients including Bank Danamon and SMBC.

SKILLS

Languages & Frameworks: Java, Spring Boot, Microservices, REST API Design, JVM, Grails, Node.js, Python, FastAPI, OpenAPI

Architecture & Patterns: Event-Driven Architecture, Design Patterns, DDD, CQRS, Event Sourcing, Clean Architecture, SOLID, TDD, Agile/Scrum, Distributed Systems

AI & Machine Learning: Agentic AI, Model Context Protocol (MCP), FastMCP, LangChain, RAG, OpenAI API, Anthropic Claude, Ollama, Qwen, Gemini, Prompt Engineering, Keras, TensorFlow, PyTorch

Databases & Vector Stores: Oracle PL/SQL (CDB/PDB), PostgreSQL, MySQL, Redis, Flyway, pgvector, ChromaDB, FAISS, PySpark, Databricks

Cloud & DevOps: AWS (EKS, ECS, Lambda, S3), GCP, Azure, Docker, Kubernetes, Apache Kafka, MQTT, VerneMQ, Nginx, CI/CD, GitHub Actions

Banking & Security: Oracle FLEXCUBE, Oracle Banking Payments (OBPM), SWIFT, OAuth2, JWT

EXPERIENCE

Lead Software Engineer | *MehvozSoft* *Aug 2025 – Present*

- Direct the design and production integration of enterprise Agentic AI pipelines and custom LLM interfaces using LangChain, RAG, and MCP.
- Designed and developed self-executing AI agents with LangChain, orchestrating multi-step reasoning to reduce manual oversight by 40%.
- Engineered RAG architectures so private language models can query internal document repositories and call external APIs via tool-calling.
- Spearhead modernization of legacy business layers into high-availability Microservices on Java, Spring Boot, Kubernetes, Kafka, and cloud infrastructure.
- Integrated resilient messaging pipelines with Kafka and Redis to streamline background processing and improve Java / Spring Boot performance.
- Optimize system stability and server performance across distributed services.

Senior Software Engineer (Java) | *Midas Software Solutions Ltd.* *Mar 2023 – Aug 2025*

- **Bank Danamon (Indonesia):** Architected and executed migration of a core digital banking monolith to a distributed Spring Boot Microservices structure, delivering 70% faster deployment cycles for high-concurrency instances.
- Designed, optimized, and maintained over 100 microservice REST endpoints using OAuth2 and JWT, supporting 50k+ concurrent users.
- **SMBC (Singapore) and Oracle:** Customized and deployed banking modules inside Oracle FLEXCUBE and Oracle Banking Payments (OBPM) using JavaScript, XML, and Oracle PL/SQL to execute cross-border SWIFT transactions.
- Applied System Design and Distributed Systems practices (Kafka, Kubernetes, Docker, CI/CD) on production banking workloads with PostgreSQL and Oracle.

Application Developer | *MehvozSoft* *Sep 2021 – Feb 2023*

- Developed and deployed cross-platform mobile utilities (including Express Entry Calculator and Adhaan applications) on iOS and Android using Ionic and Angular.
- Crafted database schemas and optimized SQL queries, indexing, and procedures on PostgreSQL and MySQL to accelerate transactional execution.

PROJECTS

Public original demos (Repos: github.com/tanvir-ux)

• **agentic-mcp-gateway** | *Python, FastAPI, MCP, RAG, Docker*
Built an MCP server plus HTTP agent (FastAPI) sharing one tool registry for document retrieval, payment searches, and ticket creation. Implemented a ReAct agent loop with CI stubbing and configurable live LLM providers.

• **banking-payments-spring** | *Java 17, Spring Boot 3.3, Kafka, PostgreSQL, Docker*
Implemented a credit-transfer REST microservice featuring JWT resource security, idempotent POST execution, JPA persistence, AML sanctions stubbing, and ISO 20022/SWIFT-inspired Kafka domain events.

• **kafka-spring-microservices** | *Java, Spring Boot, Kafka, Kubernetes, PostgreSQL*
Built three cooperating Spring Boot microservices with double-entry ledger execution, idempotent consumers, publish-after-commit event handling, and Kubernetes deployment manifests.

Open-Source Contributions

- **Meta Docusaurus:** Addressed a race condition in collapsible layout by adding programmatic bypass routes in `CollapsibleBase` (PR #12196).
- **Google Chrome / Lighthouse:** Contributed diagnostic parameters for Lighthouse auditing (Issue #10721).
- **Keras:** Authored and merged layout-logic updates in `keras-team/keras-io` (PR #596).

EDUCATION

Stamford University Bangladesh — *B.Sc. in Electrical and Electronic Engineering (EEE)*
Graduated Aug 2021 | CGPA: 3.25

Machine Learning Specialization — *Stanford University and DeepLearning.AI (Coursera)*
Credential ID: 4UGJY8KESNCZ